

Exploratory Data Analysis

Airline On-Time Performance

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CoNVO overview

- ▶ Identifying flights at high risk of delay months in advance is a key challenge for United Airlines' network planning team
- ▶ Understanding delay drivers will highlight underperforming routes and markets to inform better scheduling decisions
- ▶ A model will be built using historical flight data to predict the probability and intensity of departure delays
- ▶ Model outputs will give the network planning team an expectation of delays so candidate schedules can be adjusted to balance an aggressive schedule with operational efficiency

Dataset choice

- ▶ Airline Service Quality Performance 234 dataset from Bureau of Transportation Statistics
- ▶ This dataset reports every flight in a month, when it was scheduled to leave/arrive, and when it actually left/arrived
- ▶ Data is reported by each airline
- ▶ For the scope of this project, we'll take United Airlines data between October 2024 and September 2025

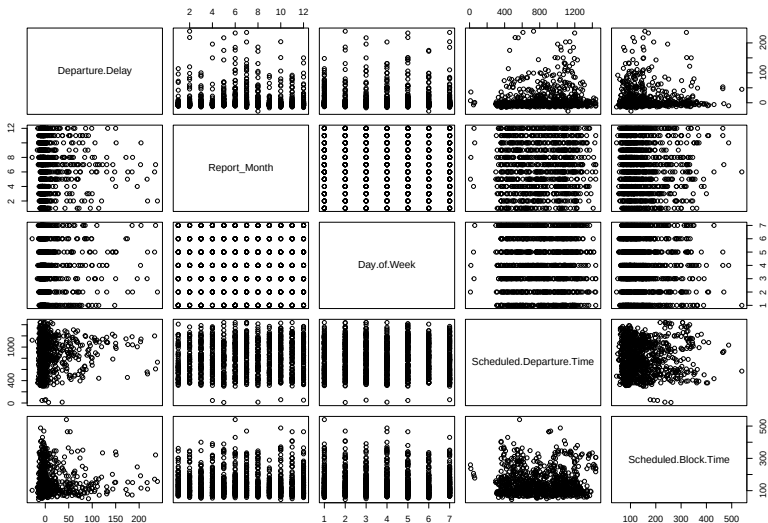
<https://www.bts.gov/browse-statistical-products-and-data/bts-publications/airline-service-quality-performance-234-time>.

Dataset info

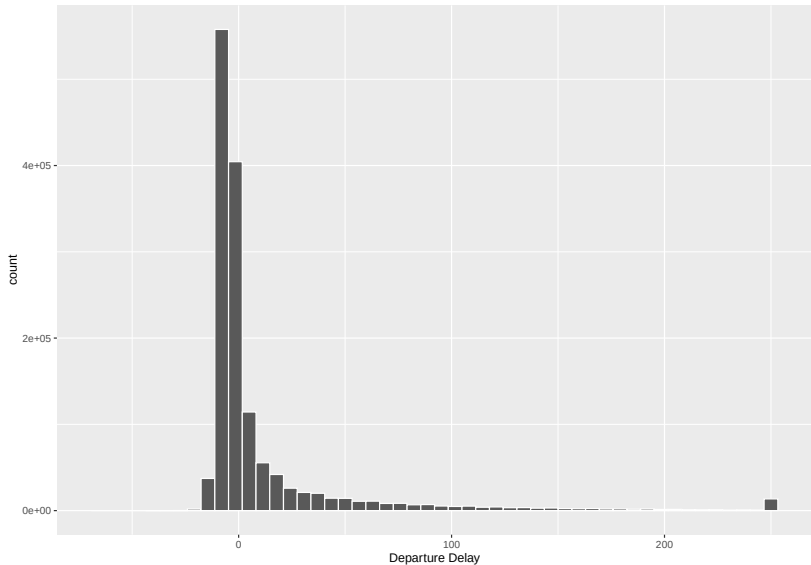
- ▶ Split out cancelled flights into a separate dataset
- ▶ 1,434,047 rows, each corresponding to a departed flight
- ▶ Relevant columns: Departure airport, Arrival airport, Day of week, Scheduled departure time, Scheduled arrival time, Scheduled block time, Report month, Departure delay, and Arrival delay

Pairs plot

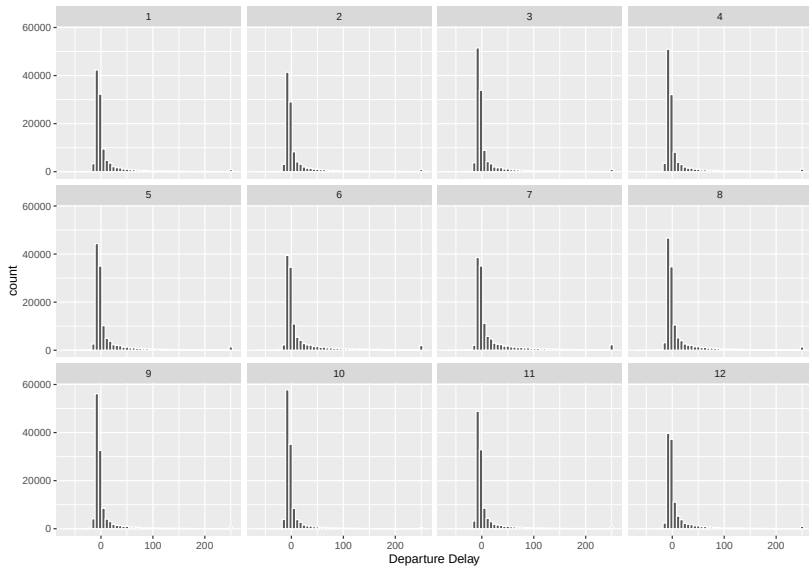
- ▶ Random sample of 1000 flights



Histogram of departure delays

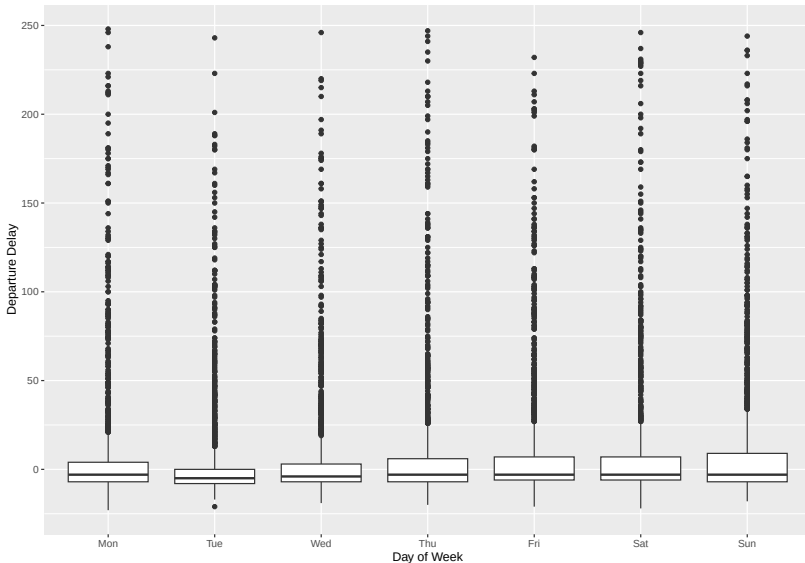


Histograms of departure delays by month



Boxplot by day of week

- ▶ Random sample of 10,000 flights
- ▶ Filtered out flights that were delayed over 250 minutes

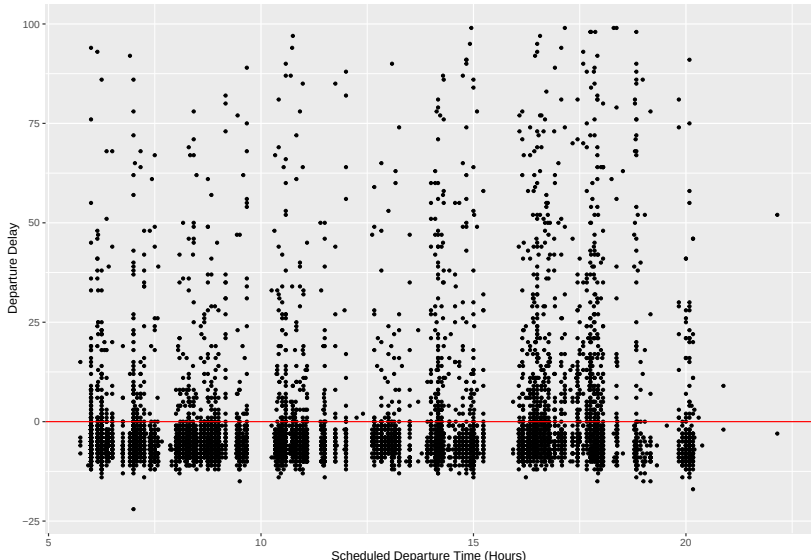


Percentiles by day of week

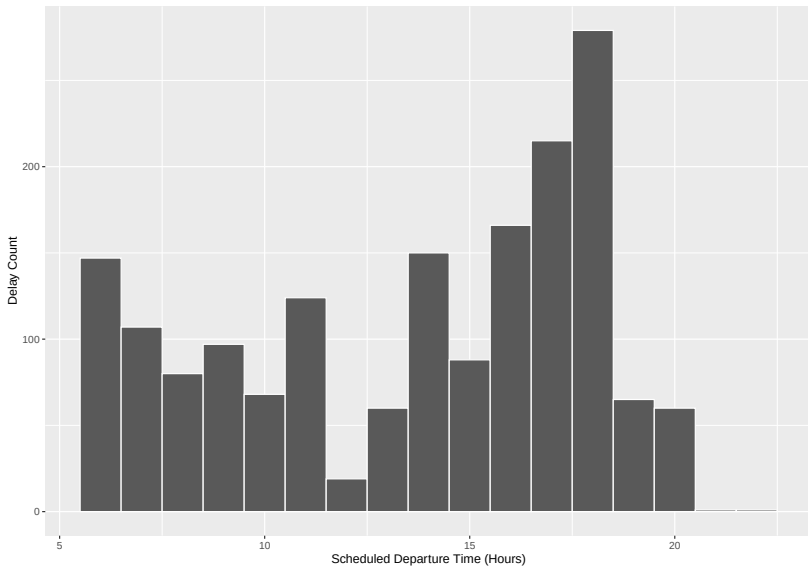
```
## # A tibble: 7 x 3
##   Day.of.Week  p75    p95
##   <fct>        <dbl> <dbl>
## 1 Mon          6    92
## 2 Tue          1  62.8
## 3 Wed          3  67.4
## 4 Thu          8  94.4
## 5 Fri          7   73
## 6 Sat          7   80
## 7 Sun          6  74.3
```

Scatterplot of departure delay vs scheduled departure time at PIT

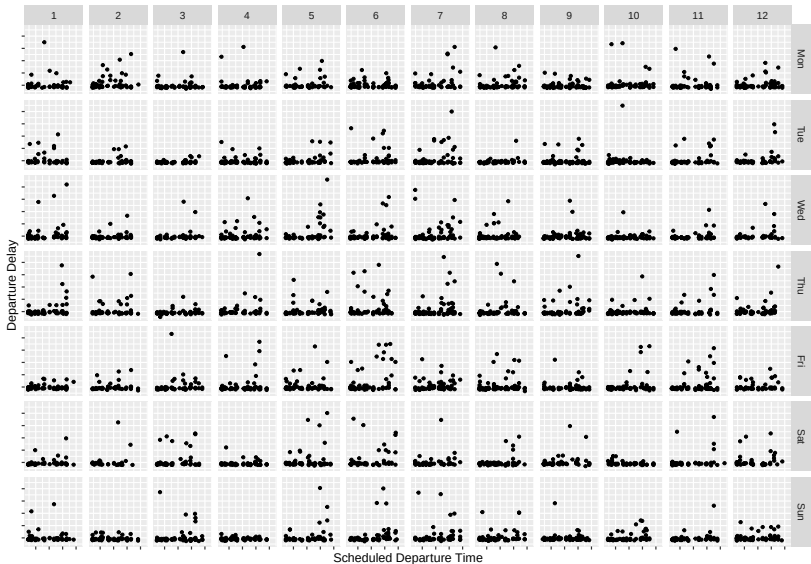
- Filtered out flights that were delayed over 100 minutes



Histogram of delays by scheduled departure time at PIT



Departure delays by month, day of week, and scheduled departure time at PIT



Causal relationships

